Journal of Guidance, Control, and Dynamics, Vol. 42, No. 3, 2019, pp. 447–457, DOI: 10.2514/1.G004048

Augmented Apollo Powered Descent Guidance

Ping Lu*
San Diego State University, San Diego, California 92182-1308

The Apollo powered descent guidance (APDG) paradigm, developed over 50 years ago for the Apollo Lunar Module, is beautifully simple and elegant. Still, the limitations of APDG have become increasingly apparent and restraining as more diverse applications are considered. This paper presents a methodology, dubbed Augmented Apollo Powered Descent Guidance (A^{2}PDG), to significantly enhance APDG. In A^{2}PDG, the user can choose from a family of *tunable* guidance laws of APDG variants by selecting a gain. This approach unifies the two versions of APDG guidance laws in the literature as they are shown to be special cases of the tunable guidance laws. Furthermore, the guidance law offers the flexibility to balance the competing considerations among fuel usage, trajectory shaping, and guidance command variations. Another prominent feature of A^{2}PDG is the onboard determination of an appropriate condition to start the powered descent phase and corresponding flight time, which helps yield near-minimum fuel consumption, ensure accurate landing, and preserve sufficient guidance margins for dispersions. Extensive end-to-end closed-loop Monte Carlo simulations for a human Mars mission are used to demonstrate the working of A^{2}PDG and its advantages.

I. Introduction

By the time this paper is published, we will be celebrating the 50th anniversary of the lunar landing of Apollo 11. The Apollo powered descent guidance (APDG) technology that landed the Apollo Lunar Module on the Moon is one of the many technological innovations of the time that enabled the ultimate success. The APDG law, as flown, assumes a thrust acceleration vector profile of a quadratic function of time. [1–3] Using the current position and velocity vectors as the initial condition, the coefficients of the quadratic polynomial are uniquely determined by meeting the targeted final position, velocity, and thrust acceleration (all vectors) at a specified final time. Another version of the powered guidance law developed at the same time is known as the E-guidance [4], although it was not actually flown. In the E-guidance approach (likely named after the notation to a key matrix in the derivation), the commanded thrust acceleration vector is defined by a linear polynomial of time, and the targeting condition is specified by the final position and velocity vectors at the prescribed final time. Later it is discovered that the E-guidance law is actually the solution to an optimal landing problem in a constant gravity field with a quadratic performance index on the thrust acceleration. [5] Even though E-guidance is not theoretically fuel optimal (which requires the performance index to be an integral of the thrust acceleration magnitude, not a quadratic performance index), it typically yields a smaller propellant consumption in the same problem than APDG for the same burn time.

Both APDG and E-guidance laws have simple and elegant closed-form solutions. Since the Apollo program, APDG has been the baseline descent guidance approach for numerous planetary landing studies and actual missions. [6–10] Still, APDG and E-guidance represent a technology of more than 50 years old. There are aspects of APDG and E-guidance that are becoming the bottleneck for the venerable technology to meet the ever increasing demand for mission autonomy and higher system performance. Two such major issues are the determination of the powered descent initiation (PDI) condition and initial time-to-go. The PDI point is where the engine is ignited and the powered descent phase is commenced. Time-to-go is a parameter in the E-guidance and APDG laws that specifies the remaining burn time. In the Apollo program, the PDI trigger is a pre-selected threshold of the ground range. [2] Most of the follow-on applications adopt a similar technique for a single-parameter threshold-based PDI trigger outside the powered descent guidance system. The importance of an appropriate PDI condition and the initial value of the time-to-go cannot be overstressed. The PDI condition is highly critical to the success of APDG and E-guidance, because it defines the divert requirement, propellant usage, and whether the required thrust magnitude will saturate severely on the maximum or minimum allowable bound (which in turn determines whether the targeting condition can be achieved.) [11] However, a good PDI condition is really a function of the full trajectory state, and its determination cannot be adequately captured by the information on a single parameter. Therefore a PDI trigger based on the threshold value of one variable/parameter either must be sufficiently conservative at the cost of a considerable propellant penalty, or risks to fail to lead to successful landing in significantly off-nominal or dispersed cases.

The value of time-to-go has similar effects. Too short or too long a time-to-go can render the powered descent

*Professor and Chair, Department of Aerospace Engineering; plu@sdsu.edu, Fellow AIAA

problem infeasible. Under otherwise the same condition, a shorter (yet still feasible) time-to-go tends to cause a smaller propellant usage. But a most appropriate initial value of time-to-go is really dependent on the PDI condition. The conventional way to determine the PDI trigger and corresponding time-to-go is through trial and error in simulations on the ground. And they will necessarily be conservative in order to accommodate dispersed cases. This means that the propellant performance is likely to suffer a substantial penalty.

Motivated by the on-going need for reliable and high-precision landing in planetary missions (robotic or human in the future) and the renewed interest in human lunar landing missions, a methodology we call Augmented Apollo Powered Descent Guidance (A^2PDG) is developed in this paper to combine the invaluable heritage of APDG and some results of the recent advanced descent guidance research toward a fully autonomous powered descent system. At the front end of A^2PDG is still the flight-proven Apollo powered guidance method, except that a *tunable* Apollo powered guidance (TAPDG) law reigns. Unlike the classical APDG or E-Guidance law where no parameter is left for the user to adjust, the TAPDG law has a guidance gain that can be tuned. The E-guidance law and APDG law turn out to be the special cases of the TAPDG law for certain values of that guidance parameter. But for other values of the gain the guidance law is a new variant that trades off between propellant usage and trajectory shaping. This tunable guidance law is further augmented by the unique capability of A^2PDG of autonomous determination of a favorable PDI condition and corresponding time-to-go. As part of its functionalities, A^2PDG determines the PDI point on the basis of the actual fight condition by a recently developed algorithm dubbed Universal Powered Guidance (UPG). [12]. UPG is an indirect-method-based algorithm that is capable of rapidly solving several variations of the fuel-optimal powered descent problem. The PDI point is found on-board such that the propellant usage required for meeting the landing condition is nearly minimum, and the divert requirement is minor. The UPG solution also provides the baseline for the initial time-to-go in the tunable guidance law. Adequate guidance margins can be conveniently built in by the user in the setting of the algorithm for determining the PDI condition in accordance with the anticipated levels of system uncertainty and trajectory dispersions.

Often times it is desired to have the vehicle touch down along a short vertical descent final phase where the velocity is held at a specified constant. To accommodate possible dispersions at the beginning of this vertical descent, a closed-loop gravity-turn guidance law is developed to regulate the velocity to ensure the satisfaction of the touchdown descent rate constraint.

The proposed A^2PDG method is applicable whether the flight before the powered descent phase is lifting entry (as in a Mars mission), or a ballistic trajectory (as in a lunar mission or ballistic planetary entry). In this paper a human Mars mission with all-propulsion descent and landing by a supersonic retropropulsion system (SRP) is used to demonstrate the key aspects of A^2PDG in extensive closed-loop Monte Carlo simulations from entry interface to touchdown.

II. Tunable Apollo Powered Descent Guidance Law

A. E-Guidance and Apollo Powered Descent Guidance

In an inertial Cartesian coordinate system, the 3-dimensional point-mass powered descent dynamics over a planet may be expressed as

$$\dot{\boldsymbol{r}} = \boldsymbol{V} \tag{1}$$

$$\dot{\boldsymbol{V}} = \boldsymbol{g} + \boldsymbol{a}_T \tag{2}$$

where $\boldsymbol{r} \in R^3$ and $\boldsymbol{V} \in R^3$ are the position and velocity vector of the vehicle, and $\boldsymbol{g}$ the gravitational acceleration vector which is treated as a constant (although not necessarily. See also Ref. [12]). The thrust *acceleration* vector is $\boldsymbol{a}_T$. It is assumed that either the atmosphere is absent or aerodynamic forces can be safely ignored in the guidance solution. Let $\boldsymbol{r}_f^*$ and $\boldsymbol{V}_f^*$ be the prescribed position and velocity vectors at a specified final time t_f that define the coordinates of the landing site and required touchdown velocity (direction and magnitude). The targeting condition is

$$\boldsymbol{r}(t_f) = \boldsymbol{r}_f^* \tag{3}$$

$$\boldsymbol{V}(t_f) = \boldsymbol{V}_f^* \tag{4}$$

The current position and velocity vectors $\boldsymbol{r}(t)$ and $\boldsymbol{V}(t)$ are known. The guidance problem is to find the thrust acceleration command $\boldsymbol{a}_T(t)$ to steer the vehicle from the current state to the specified targeting condition $(\boldsymbol{r}_f^*, \boldsymbol{V}_f^*)$ at the required final time t_f.

In the E-guidance approach a linear function of time profile is assumed for $\boldsymbol{a}_T$

$$\boldsymbol{a}_T = \boldsymbol{c}_0 + \boldsymbol{c}_1\tau, \quad \tau \in [t, t_f] \tag{5}$$

where $\boldsymbol{c}_0 \in R^3$ and $\boldsymbol{c}_1 \in R^3$ are constants to be determined. Integrating Eqs. (1) and (2) with the above linear $\boldsymbol{a}_T$ and the initial condition $[\boldsymbol{r}(t), \boldsymbol{V}(t)]$, we can find the required $\boldsymbol{c}_0$ and $\boldsymbol{c}_1$ by satisfying the terminal condition in Eqs. (3) and (4). Substituting the expressions of $\boldsymbol{c}_0$ and $\boldsymbol{c}_1$ back into Eq. (5), we have the E-guidance law [4]:

$$\boldsymbol{a}_T(t) = -\frac{2}{t_{go}}\left[\boldsymbol{V}_f^* - \boldsymbol{V}(t)\right] + \frac{6}{t_{go}^2}\left[\boldsymbol{r}_f^* - \boldsymbol{r}(t) - \boldsymbol{V}(t)t_{go}\right] - \boldsymbol{g} \tag{6}$$

where the time-to-go t_{go} is defined as

$$t_{go} = t_f - t \tag{7}$$

Note that Eq. (6), or any other variant as will be seen in this and next subsections, defines both the magnitude and direction of the commanded thrust acceleration vector. It is shown in Ref. [5] that the optimal thrust acceleration profile is linear as in Eq. (5) for the optimal control problem when $\boldsymbol{g}$ is constant and the performance index is

$$J_1 = \int_t^{t_f} \boldsymbol{a}_T^T \boldsymbol{a}_T d\tau = \int_t^{t_f} \|\boldsymbol{a}_T\|^2 d\tau \tag{8}$$

With the current state as the initial condition and final condition in Eqs. (3) and (4), the optimal thrust acceleration is precisely Eq. (6). Even though the E-guidance law is not exactly fuel optimal (which requires the performance index to be $\int_t^{t_f} \|\boldsymbol{a}_T\| d\tau$), its propellant usage is close to that of the fuel-optimal solution in many cases, as has long been observed. [1, 4] This observation, however, must be qualified by the condition that t_{go} is sufficiently short, because examples can be readily found that demonstrate significant propellant advantage by the fuel-optimal solution over E-guidance for longer t_{go}.

If a quadratic thrust acceleration profile is assumed instead,

$$\boldsymbol{a}_T = \boldsymbol{c}_0 + \boldsymbol{c}_1\tau + \boldsymbol{c}_2\tau^2, \quad \tau \in [t, t_f] \tag{9}$$

and the final thrust acceleration vector is also constrained in addition to those on the final position and velocity in Eqs. (3) and (4)

$$\boldsymbol{a}_T(t_f) = \boldsymbol{a}_{T_f}^* \tag{10}$$

where $\boldsymbol{a}_{T_f}^*$ is a specified thrust acceleration vector, a similar process will lead to the APDG law [1, 3, 4]

$$\boldsymbol{a}_T(t) = -\frac{6}{t_{go}}\left[\boldsymbol{V}_f^* - \boldsymbol{V}(t)\right] + \frac{12}{t_{go}^2}\left[\boldsymbol{r}_f^* - \boldsymbol{r}(t) - \boldsymbol{V}(t)t_{go}\right] + \boldsymbol{a}_{T_f}^* \tag{11}$$

Under identical conditions, the APDG law typically consumes more propellant than the E-guidance law. But the APDG law constrains the final thrust to align in the specified direction, and this effectively controls the final body attitude of the vehicle in a desirable orientation.

For any vehicle, the thrust magnitude of the engine will have an upper bound T_{max}. On the other hand the engine cannot be completely shutdown (thrust=0) during the powered descent as the engine requires some time to ramp up the thrust from a restart. Thus there will be a non-zero minimum thrust (throttle) T_{min} the above which the thrust must be kept. Equivalently at each instant the thrust acceleration magnitude is bounded by

$$a_{T_{min}}(t) \le \|\boldsymbol{a}_T(t)\| \le a_{T_{max}}(t) \tag{12}$$

The E-guidance and APDG laws do not consider the thrust bounds in their design. Therefore the thrust magnitude will be saturated at the bound if the commanded thrust (acceleration) exceeds what the engine can provide (on the high or low side).

Both the E-guidance law and APDG law have a singularity at $t_{go} = 0$. To avoid the singularity, the update of the thrust acceleration command from Eq. (6) or (11) may be stopped when t_{go} is smaller than a prescribed value (e. g., 0.5 sec). A constant thrust acceleration command in this small final period will still provide practically good guidance accuracy.

B. Tunable Guidance Law

In Refs. [1, 3] an implicit guidance[a] version of the powered descent law is discussed. In this context, the thrust acceleration vector command is generated by tracking prescribed profiles for the position and velocity vectors:

$$\boldsymbol{a}_T(t) = \boldsymbol{a}_d(t) - \frac{k_V}{t_{go}}\left[\boldsymbol{V}(t) - \boldsymbol{V}_d(t)\right] - \frac{k_r}{t_{go}^2}\left[\boldsymbol{r}(t) - \boldsymbol{r}_d(t)\right] \tag{13}$$

[a] In the early aerospace guidance literature, "implicit guidance" refers to the approaches where the guidance command is generated based on a given reference, typically involving linearization; "explicit guidance" on the other hand implies a method that explicitly enforces the exact targeting condition and other constraints in the guidance solution in each guidance cycle, thus does not use a fixed reference trajectory.

where k_V and k_r are guidance gains, and the vectors with the subscript "d" are the desired profiles of thrust acceleration, velocity, and position vectors. In Refs. [1, 3] it is shown that for a quadratic function of $\boldsymbol{a}_d$, and the corresponding cubic profile of $\boldsymbol{V}_d$ and quartic profile of $\boldsymbol{r}_d$ that satisfy the terminal constraints in Eqs. (3), (4) and (10), a particular set of guidance gains of $k_V = 6$ and $k_r = 12$ in Eq. (13) yields exactly the APDG law in Eq. (11).

In this paper we will start with a *linear* desired thrust acceleration profile

$$\boldsymbol{a}_d = \boldsymbol{a}_{T_f}^* + \boldsymbol{n} t_{go} \tag{14}$$

where $\boldsymbol{a}_{T_f}^*$ is a given final thrust acceleration, and $\boldsymbol{n} \in R^3$ is clearly the negative of the jerk vector. The corresponding $\boldsymbol{V}_d$ from Eq. (2) (with a constant $\boldsymbol{g}$) and $\boldsymbol{r}_d$ from Eq. (1) are

$$\boldsymbol{V}_d = \boldsymbol{V}_f^* - \boldsymbol{a}_{T_f}^* t_{go} - \frac{1}{2}\boldsymbol{n} t_{go}^2 - \boldsymbol{g} t_{go} \tag{15}$$

$$\boldsymbol{r}_d = \boldsymbol{r}_f^* - \boldsymbol{V}_f^* t_{go} + \frac{1}{2}\boldsymbol{a}_{T_f}^* t_{go}^2 + \frac{1}{6}\boldsymbol{n} t_{go}^3 + \frac{1}{2}\boldsymbol{g} t_{go}^2 \tag{16}$$

where the final constraints (at $t_{go} = 0$) in Eqs. (3) and (4) have been used in the expressions of $\boldsymbol{V}_d$ and $\boldsymbol{r}_d$. All the coefficients in these desired profiles except for $\boldsymbol{n}$ are already specified. It is not immediately clear what choice should be made for $\boldsymbol{n}$ and its effect on the trajectory is also less intuitive. To rid us of this ambiguity, when these expressions of $\boldsymbol{a}_d$, $\boldsymbol{V}_d$ and $\boldsymbol{r}_d$ are substituted into the implicit guidance law Eq. (13), it can be readily shown that if k_V and k_r are chosen according to the following relationship

$$k_V = \frac{1}{3}k_r + 2 \tag{17}$$

the coefficient of $\boldsymbol{n}$ vanishes. The resulting guidance law is independent of $\boldsymbol{n}$, and contains a single gain k_r that can be adjusted. After rearranging the terms in the guidance law in a form comparable to Eqs. (6) and (11), a *tunable* guidance law emerges:

$$\boldsymbol{a}_T = \frac{2}{t_{go}}\left(1 - \frac{1}{3}k_r\right)\left[\boldsymbol{V}_f^* - \boldsymbol{V}(t)\right] + \frac{k_r}{t_{go}^2}\left[\boldsymbol{r}_f^* - \boldsymbol{r}(t) - \boldsymbol{V}(t) t_{go}\right] + \frac{1}{6}\left(k_r - 6\right)\boldsymbol{a}_{T_f}^* + \frac{1}{6}\left(k_r - 12\right)\boldsymbol{g} \tag{18}$$

Discussions:

1. When $k_r = 6$ ($k_V = 4$ by Eq. (17)), $\boldsymbol{a}_{T_f}^*$ drops out of the guidance law in Eq. (18), and the E-guidance law as in Eq. (6) ensues. While this has not been reported in the literature, it is a parallel to the finding in Refs. [1, 3] on applying Eq. (13) to a desired quadratic thrust acceleration reference and the corresponding $\boldsymbol{V}_d$ and $\boldsymbol{r}_d$, and is not a big surprise.

2. When $k_r = 12$ ($k_V = 6$ by Eq. (17)), the tunable guidance law in Eq. (18) is independent of $\boldsymbol{g}$, and Eq. (18) gives rise to exactly the APDG law in Eq. (11)! This is particularly interesting and noteworthy. Recall that Refs. [1, 3] show that the APDG law is a special case of the implicit guidance law in Eq. (13) where the desired profiles are generated by a quadratic acceleration function. Here we reveal that it can also be obtained from the implicit guidance law with the desired profiles generated by a linear acceleration function, with the same set of feedback gains!

3. Compared to the E-guidance and APDG laws, the tunable guidance law in Eq. (18) has a subtly different interpretation when $k_r \neq 6$ and $k_r \neq 12$. The E-guidance and APDG laws are derived to meet the boundary conditions at both the current time t and final time t_f (thus are *explicit* guidance laws). The tunable guidance law with $6 < k_r < 12$ is a trajectory control law to track the reference profiles $\boldsymbol{V}_d$ and $\boldsymbol{r}_d$, therefore an *implicit* guidance law. Note that $\boldsymbol{V}_d$ and $\boldsymbol{r}_d$ meet the targeting condition in Eqs. (3) and (4) at t_f, but do not necessarily pass through the current state ($\boldsymbol{r}(t)$, $\boldsymbol{V}(t)$). This means that there may be tracking errors at the current condition. Our experience has been that the terminal condition in Eqs. (3) and (4) can be achieved sufficiently accurately with any $k_r \in [6, 12]$.

4. The final acceleration condition is in general different from $\boldsymbol{a}_{T_f}^*$, unless $k_r = 12$. In fact, as $t_{go} \to 0$ and $\boldsymbol{r} \to \boldsymbol{r}_f^*$ and $\boldsymbol{V} \to \boldsymbol{V}_f^*$, the thrust acceleration command from Eq. (18) approaches

$$\boldsymbol{a}_T \to \left(\frac{1}{6}k_r - 1\right)\boldsymbol{a}_{T_f}^* + \left(\frac{1}{6}k_r - 2\right)\boldsymbol{g} \tag{19}$$

For descent and landing the choice of $\boldsymbol{a}^*_{T_f}$ is typically $\boldsymbol{a}^*_{T_f} = -\kappa \boldsymbol{g}$ where $\kappa > 1$. Thus the final thrust acceleration command is

$$\boldsymbol{a}_{T_f} = -\left[(\kappa - 1)\left(\frac{1}{6}k_r - 1\right) + 1\right]\boldsymbol{g} \tag{20}$$

For instance for $\kappa = 2.0$ and $k_r = 9.0$, $\boldsymbol{a}_{T_f} = -1.5\boldsymbol{g} = 0.75\boldsymbol{a}^*_{T_f}$. From Eq. (19), $\boldsymbol{a}_{T_f} = \boldsymbol{a}^*_{T_f}$ only when $k_r = 12$ (i. e., the APDG law). However, for a specified k_r, one can always choose the value of κ in defining $\boldsymbol{a}^*_{T_f}$ so that Eq. (20) gives the desired final thrust acceleration magnitude.

5. Let $\Delta\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{r}_d$. The closed-loop tracking error dynamics under the guidance law in Eq. (18) are

$$\Delta\ddot{\boldsymbol{r}} + \frac{1}{t_{go}}\left(\frac{1}{3}k_r + 2\right)\Delta\dot{\boldsymbol{r}} + \frac{1}{t^2_{go}}k_r\Delta\boldsymbol{r} = 0 \tag{21}$$

In the following, the discussion is for a frozen t_{go}. The characteristic equation of the closed-loop system is then

$$\lambda^2 + \frac{1}{t_{go}}\left(\frac{1}{3}k_r + 2\right)\lambda + \frac{1}{t^2_{go}}k_r = 0 \tag{22}$$

Hence the natural frequency satisfies $\omega_n t_{go} = \sqrt{k_r}$ and the damping ratio is given by

$$\zeta = \frac{1}{2\sqrt{k_r}}\left(\frac{1}{3}k_r + 2\right) \tag{23}$$

From Eq. (23) it can be derived that the minimum damping of the closed-loop system is achieved at $k_r = 6$, which yields $\zeta = \sqrt{2/3} = 0.816$. In other words, under the tunable APDG law, the minimum damping of the closed-loop system is realized by the E-guidance law.

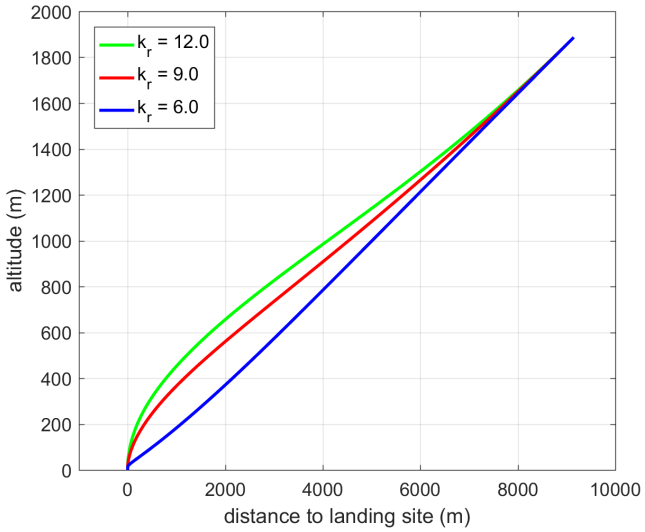
6. Under the same PDI and targeting conditions for the same vehicle, the tunable guidance law with the same t_{go} but a smaller k_r (within the range of [6, 12]) uses less propellant, but the powered descent trajectory is shallower. Thus tuning the guidance gain k_r in Eq. (18) allows the flexibility of trade off between propellant usage and trajectory shaping. In addition, if a vertical landing phase is required, the transition from the powered descent to the vertical landing phase is more aggressive with a smaller k_r. A less nimble vehicle will likely necessitate the use of a larger k_r so that the guidance commands is less aggressive and can be satisfactorily tracked by the control system. Figure 1 shows 3 powered descent trajectories with different values of k_r in a high-mass Mars landing mission with SRP (See Section V for the details). For those cases $\boldsymbol{a}^*_{T_f} = -2\boldsymbol{g}$ is used. The effect of k_r on the descent trajectory is clear and significant. The propellant usage with $k_r = 6.0$ is 99 kg (1.2%) than that with $k_r = 9.0$ which is in turn 126 kg (1.5%) less than that with $k_r = 12.0$. It should be mentioned all 3 trajectories end with a guided vertical landing phase of 5 seconds (see Fig. 1-(b)) which is the subject of the next section. In the 3DOF simulations the control authority limitations of the vehicle are not considered, thus the transition to the vertical landing phase is always achieved with all the 3 values of k_r. The beginning of (or transition into) the vertical descent phase is marked by the condition that $t_{go} = 0$ in the guidance law in Eq. (18).

III. Descent Rate Regulation Guidance Law for Gravity Turn

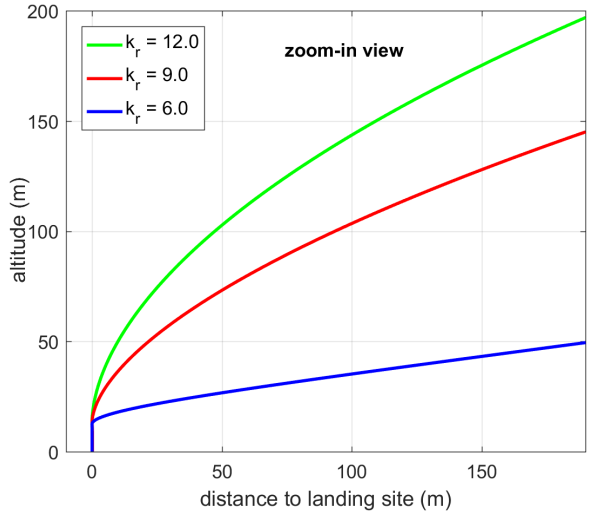
An all-propulsive descent mission typically ends with a vertical landing phase where the descent rate is kept at a constant. The guidance system must be prepared for the cases where at the start of this landing phase, the trajectory is not exactly in the vertical direction, and the altitude and velocity may all be different from nominal values. A gravity-turn trajectory where the thrust is pointed to the negative direction of the velocity vector is most natural and fuel efficient to bring the trajectory into vertical descent (see Fig. 2). The magnitude of the thrust may be modulated to maintain or control the descent rate. To adapt to all possible dispersions at the start of this landing phase and still land the vehicle along the gravity-turn trajectory with the specified touchdown descent rate, a closed-loop descent-rate regulation law is developed in this section for gravity turn.

Let h and $\dot{h}$ be the altitude and altitude rate at the current time t, and h^*_f and $\dot{h}^*_f$ be the specified touchdown altitude and altitude rate (note that "descent rate" is the negative of the altitude rate). We seek to determine the required thrust acceleration in the gravity turn so that the final altitude rate is equal to $\dot{h}^*_f$ when the altitude reaches h^*_f. The simplest option is to use a *constant* vertical acceleration. Let this constant vertical acceleration be a_h. Note that in the following derivation t is fixed. For $\tau \geq t$, we have

$$\dot{h}(\tau) \quad = \quad \dot{h}(t) + a_h(\tau - t) \tag{24}$$



a) Descent trajectories under tunable powered descent guidance law



b) Zoom-in view of the descent trajectories under tunable powered descent guidance law

Figure 1. The effects of the gain k_r in the tunable powered descent guidance law in Eq. (18)

$$h(\tau) = h(t) + \dot{h}(t)(\tau - t) + \frac{1}{2}a_h(\tau - t)^2 \tag{25}$$

Let $\tau_f > t$ be the time when $h(\tau_f) = h_f^*$, and denote $\tau_{go} = \tau_f - t$. Set $h(\tau_f) = h_f^*$ and $\dot{h}(\tau_f) = \dot{h}_f^*$ in the above equations

$$\dot{h}(t) + a_h\tau_{go} = \dot{h}_f^* \tag{26}$$

$$h(t) + \dot{h}(t)\tau_{go} + \frac{1}{2}a_h\tau_{go}^2 = h_f^* \tag{27}$$

The solutions for a_h and τ_{go} are

$$a_h = \frac{\dot{h}_f^{*2} - \dot{h}^2}{2(h_f^* - h)} \tag{28}$$

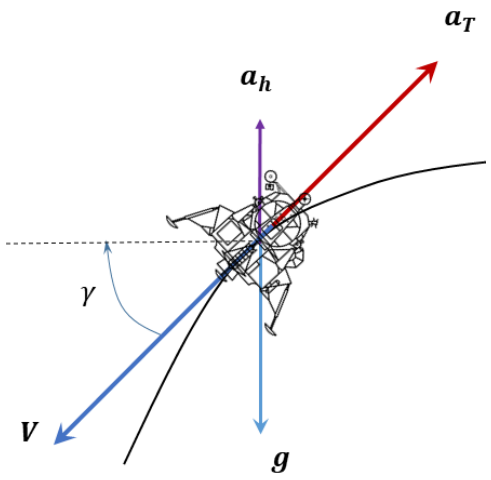


Figure 2. Gravity-turn descent

$$\tau_{go} = \frac{2(h_f^* - h)}{\dot{h}_f^* + \dot{h}} \tag{29}$$

where the argument of t for h and $\dot{h}$ has been dropped for brevity. The required thrust acceleration magnitude to provide the vertical acceleration in Eq. (28) can be computed in reference to the geometry in Fig. 2 as follows:

$$a_h = a_T \sin(-\gamma) - g \tag{30}$$

where γ is the flight path angle (and is negative as shown in Fig. 2). Therefore

$$a_T = -\frac{a_h + g}{\sin\gamma} \tag{31}$$

Define the unit vector in the direction of the velocity vector

$$\mathbf{1}_V = \frac{\boldsymbol{V}}{\|\boldsymbol{V}\|}$$

The gravity-turn guidance law for the thrust acceleration vector that regulates the descent rate is then

$$\boldsymbol{a}_T = -a_T \mathbf{1}_V = \frac{a_h + g}{\sin\gamma}\mathbf{1}_V = \frac{1}{\sin\gamma}\left[\frac{\dot{h}_f^{*2} - \dot{h}^2}{2(h_f^* - h)} + g\right]\mathbf{1}_V \tag{32}$$

If at the start the vehicle is already in vertical descent with the required descent rate ($\gamma = -90$ deg and $\dot{h} = \dot{h}_f^*$), then the guidance law in Eq. (32) simply maintains a constant-velocity vertical descent. But the guidance law is capable of taking out initial condition dispersions and landing the vehicle with the required touchdown descent rate even if the starting condition is not exactly in vertical descent. The current estimated time-to-go to touchdown is from Eq. (29).

$$t_{touchdown} = \frac{2(h_f^* - h)}{\dot{h}_f^* + \dot{h}} \tag{33}$$

It can be shown by using the L'Hospital's rule that when $\dot{h} \to \dot{h}_f^*$ and $h \to h_f^*$, the term $(\dot{h}_f^{*2} - \dot{h}^2)/2(h_f^* - h)$ in Eq. (32) is finite. Thus theoretically there is no singularity at $h = h_f^*$ when the descent rate is well regulated as desired. But to avoid singularity caused by any descent rate regulation error (for instance, error due to navigation uncertainty), the update of the guidance command from Eq. (32) may be stopped after the altitude is within a threshold value to h_f^*.

One practical issue in using the guidance law in Eq. (32) in low velocity (e. g., a few meters per second) is that the direction of $\mathbf{1}_V$ can change rapidly if there is any imperfection of pointing the thrust vector exactly along $-\mathbf{1}_V$ (for instance, due to navigation uncertainty or control system delays). Instability can arise as a result when the guidance

law keeps chasing the rapidly varying direction of $-\mathbf{1}_V$. Let $\mathbf{1}_r$ be the unit vector in the vertical direction (positive up). If the trajectory has already a negative flight path angle of magnitude close to -90 deg at the beginning of such a low-velocity gravity turn, a much more stable alternative to guidance law (32) in low-velocity applications is

$$\boldsymbol{a}_T = \frac{1}{(-\sin\gamma)}\left[\frac{\dot{h}_f^{*2} - \dot{h}^2}{2(h_f^* - h)} + g\right]\mathbf{1}_r \approx \left[\frac{\dot{h}_f^{*2} - \dot{h}^2}{2(h_f^* - h)} + g\right]\mathbf{1}_r \tag{34}$$

IV. Onboard Powered Descent Initiation Determination

A. Review of Universal Powered Guidance

The UPG algorithm developed in Ref. [12] is briefly reviewed here as it plays the central role in our PDI strategy. The problem formulation in UPG in a Cartesian coordinate system with the origin at the center of mass of the planet starts with the three-dimensional point-mass equations of motion for a rocket-powered vehicle

$$\dot{\boldsymbol{r}} = \boldsymbol{V} \tag{35}$$

$$\dot{\boldsymbol{V}} = \boldsymbol{g} + \frac{\boldsymbol{T}(t)}{m(t)} \tag{36}$$

$$\dot{m} = -\frac{\|\boldsymbol{T}\|}{v_{ex}} \tag{37}$$

$$T_{min} \leq \|\boldsymbol{T}(t)\| \leq T_{max} \tag{38}$$

where $\boldsymbol{T}$ is the thrust vector, and m the mass of the vehicle. The effective exhaust velocity of the rocket engine v_{ex} is considered a constant. The lower thrust bound T_{min} and upper bound T_{max} are constant. The powered descent problem in UPG is still to determine the thrust vector to meet the specified targeting condition. The main distinctions in UPG as compared to APDG are

1. The thrust magnitude constraints in Eq. (38) are explicitly taken into account in the guidance solution.

2. While the terminal velocity constraints in Eq. (4) remain, different terminal constraints on $\boldsymbol{r}(t_f)$ than Eq. (3) are allowed (see for instance Eq. (40) later).

3. A performance index is minimized

$$J_2 = \kappa\left[\boldsymbol{r}(t_f) - \boldsymbol{r}_f^*\right]^T\left[\boldsymbol{r}(t_f) - \boldsymbol{r}_f^*\right] + \int_0^{t_f}\frac{\|\boldsymbol{T}(t)\|}{v_{ex}}dt \tag{39}$$

   where $\kappa \geq 0$ is a constant. The integral term represents the propellant consumption. When $\kappa = 0$, this is a pure fuel-optimal problem.

4. The powered descent time t_f is not fixed *a priori*, but obtained as a part of the optimal solution

In Ref. [12], the problem is called the pinpoint landing problem when $\kappa = 0$ and the terminal constraints are exactly Eqs. (3) and (4). For the PDI determination purpose, the solution of the *soft-landing* problem [12] is used where $\kappa = 0$ and the terminal constraints are

$$\|\boldsymbol{r}(t_f)\| = \|\boldsymbol{r}_f^*\| \tag{40}$$

$$\boldsymbol{V}(t_f) = \boldsymbol{V}_f^* \tag{41}$$

In the soft-landing problem, the fuel-optimal solution is sought that lands the vehicle on the ground (Eq. (40), assuming $\|\boldsymbol{r}_f^*\|$ is the radius of the spherical planet) with the required touchdown velocity. The landing location is however not constrained. It is shown in Ref. [12] that the soft-landing problem uses the least amount of propellant among the several problems discussed therein.

The UPG algorithm in Ref. [12] numerically solves the fuel-optimal powered descent problem based on the necessary conditions as a root-finding problem of 7 variables, but with analytical expressions of the functions involved. A step-size controlled (damped) Newton-Raphson method works well. The solution process of the soft-landing problem is the most robust and fastest when compared to other variants of the optimal powered descent problem.

B. Onboard PDI Determination

An adaptive PDI determination approach is proposed in conjunction with the deployment of UPG in Ref. [12, 13] mainly for further reducing propellant usage in a Mars EDL mission. The same logic, with an important caveat to be discussed in the next subsection, remains applicable in planetary EDL applications when a version of the Apollo powered descent guidance law is employed.

Let us state the underlying assumption required for this strategy as follows.

Assumption

Prior to the powered descent phase, the vehicle is on a ballistic, lifting entry, or any other combinations that are independent of the powered descent trajectory. The only requirement is that this trajectory leads toward the general direction of the landing site, so that the range-to-go to the landing site is decreasing along the trajectory (at least until some point near the end of the trajectory).

This assumption covers descent and landing on the Moon and other atmosphere-less planets, or ballistic entry over a planet with atmosphere, as well as actively guided lifting entry over a celestial body with atmosphere (e.g, Mars). Note that the statements of the assumption are intentionally less precise because the specific nature of this trajectory is immaterial, and the trajectory prior to powered descent need not be subject to a particular accuracy condition. Moreover, this trajectory can be different depending on the actual condition on the day of the mission (that is, dispersions are allowed in this pre-powered-descent trajectory). This assumption certainly is met in any well-planned mission.

The onboard PDI logic makes use of the powerful and reliable capability of UPG in finding the fuel-optimal soft-landing solution and the complete associated trajectory information, starting at any point along the incoming trajectory. In particular, the touchdown location along the soft-landing trajectory will be predicted, and the range traveled along the soft-landing trajectory will be known. This predicted touchdown range initially will be shorter than the current ground range-to-go to the landing site. See Fig. 3. As the vehicle continues its flight on the unpowered trajectory, the assumption ensures that the actual range-to-go will be decreasing. Eventually, the predicted soft-landing range will exceed the current range-to-go as the vehicle approaches the landing site on the unpowered flight, as illustrated by the latter soft-landing trajectory in Fig. 3. This is the time when PDI should be commenced.

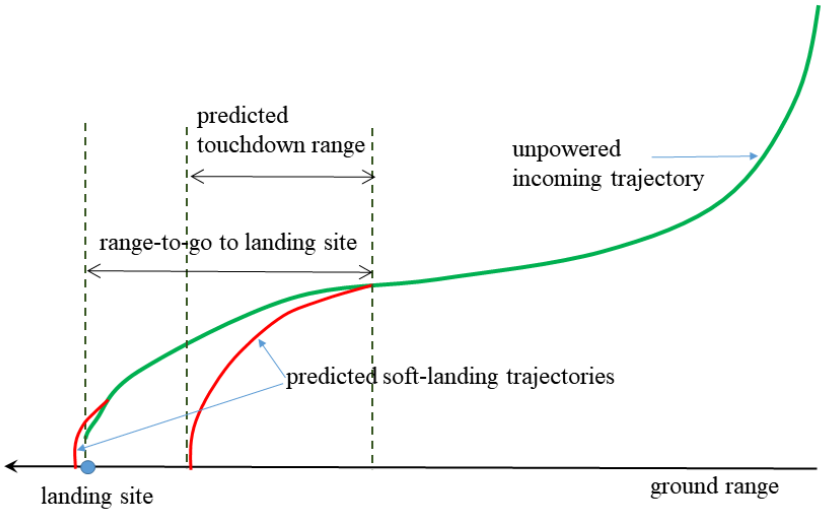


Figure 3. Predicted soft-landing trajectories along the unpowered incoming trajectory

Recall that the soft-landing trajectory demands no divert maneuver from the vehicle and the propellant consumption is the least among all powered descent trajectories that land softly on the ground. The assumption also implies that the soft-landing touchdown location from the PDI found in the above way will not be unreasonably far from the designated landing site. Thus the divert required to land at the designated site will be moderate, and the propellant usage will be close to optimal.

Note that in the PDI logic described above, the comparison of ground ranges is used. One might be tempted to use the distance between the predicted soft-landing touchdown location and the designated landing site as the trigger for PDI: when this distance becomes minimum, the PDI starts. The difficulty is that the minimum of this distance is not known in advance. And there is even no guarantee that the variations of this distance are unimodal, that is, it

decreases monotonically first, then increases monotonically after it reaches the minimum. Therefore, monitoring this distance and triggering PDI as soon as this distance begins to increase can end up starting PDI prematurely as we have observed in our simulations. In contrast, checking the sign of the difference between the predicted soft-landing touchdown range and actual range-to-go is not predicated on any particular assumed variation pattern, and this sign is assured to change as the vehicle continues to fly toward the landing site.

The following pseudo code summarizes the PDI determination algorithm. This code only needs to be executed (onboard) after the vehicle gets sufficiently close to the landing site, typically defined by a threshold value in velocity (if the unpowered flight is entry flight) or ground range (if the unpowered flight is ballistic vacuum flight). The PDI algorithm can be called at a relatively low frequency (e. g., 1 Hz or lower) and still achieves the intended objective. The 3rd step in the algorithm is a fail-safe insurance against the unlikely event where Step 2 of the algorithm never triggers the PDI, a situation we have not experienced in the extensive testing we have done so far.

Algorithm 1: Powered descent initiation determination

1: Call the soft-landing mode of UPG, and find the touchdown range along the predicted soft-landing trajectory;
2: If the touchdown range is longer than the current range-to-go, save the predicted powered descent time t^*_{SL} of the soft-landing trajectory, exit and commence PDI;
3: If the current range-to-go is shorter than a pre-specified threshold, exit and commence PDI;
4: Continue on the unpowered flight; repeat from Step 1 in the next cycle.

Once the PDI has been identified, in the A^{2}PDG framework the tunable Apollo guidance law in Eq. (18) will be used in the ensuing powered descent. The initial value of t_{go} is set by

$$t_{go} = \eta t^*_{SL} \tag{42}$$

where, as defined in the pseudo code, t^*_{SL} is the optimal time of flight of the final soft-landing trajectory found in the PDI logic. The scalar $\eta > 1.0$ is a relaxation coefficient to account for the fact that the required powered descent time is likely to be slightly longer than t^*_{SL} because of the inevitable divert maneuver to the landing site (which is not present in the soft-landing trajectory). An η of 1.2 typically suffices for all $k_r \in [6.0, 12.0]$ in the guidance law in Eq. (18). The subsequent value of t_{go} is obtained by subtracting from this initial value in Eq. (42) the time since PDI. The guidance performance is found not to be sensitive to the value of η.

This PDI determination logic does not rely on the specific nature of the unpowered trajectory prior to powered descent. Therefore it is applicable to both Mars and lunar landing, as well as other planets.

C. Guidance Margin Guarantee

It is theoretically established in Ref. [12] that the free-time fuel-optimal powered descent solution in a constant gravity field in general has a 3-arc bang-bang thrust magnitude profile of T_{max}-T_{min}-T_{max}. See the profile in red solid line in Fig. 4. The burn times of the 2nd and 3rd arc are optimized by UPG. See Ref. [12] for more details. For a given unpowered trajectory, the onboard PDI logic will enable the powered descent guidance algorithm to use nearly the minimum amount of propellant. However, if the unpowered trajectory changes, the PDI condition will change, hence the associated minimum propellant usage. It is discovered in Refs. [11, 13] that when the unpowered flight is entry flight (lifting or ballistic), the best unpowered trajectory that renders the overall least propellant usage is always one where the UPG soft-landing solution that defines the PDI has zero time duration for the throttle-back phase, that is, $\Delta t_2 = 0$ in Fig. 4. This is because in this case the PDI point is pushed as late as possible along the entry trajectory so that the PDI velocity is at the lowest value before it is too late to start powered descent. The latest possible PDI point is one where the vehicle can land only with the engine at the full throttle all the way (thus the arc at T_{min} has a zero duration).

For any member of the Apollo-class descent guidance law family as in Eq. (18), such a PDI condition (that requires full thrust to land) leaves no margin on the thrust magnitude – recall that an Apollo-class guidance law modulates both the magnitude and direction of the thrust vector. It is most likely that large errors in both landing location and touchdown velocity will occur as a consequence. A simple and effective measure is needed to ensure a guaranteed degree of margin in the PDI determination if the A^{2}PDG method is to be useful.

The simplest technique is to impose a nonzero lower bound on Δt_2 (cf. Fig. 4) in the UPG algorithm for the soft-landing solution in the PDI algorithm:

$$0 < \Delta_2 \le \Delta t_2 \tag{43}$$

where $\Delta_2 > 0$ is a user-selected constant. This constraint prevents Δt_2 from shrinking to zero in the optimal solution, as it often does otherwise. From a thrust impulse (the time integration of thrust magnitude) perspective, a nonzero

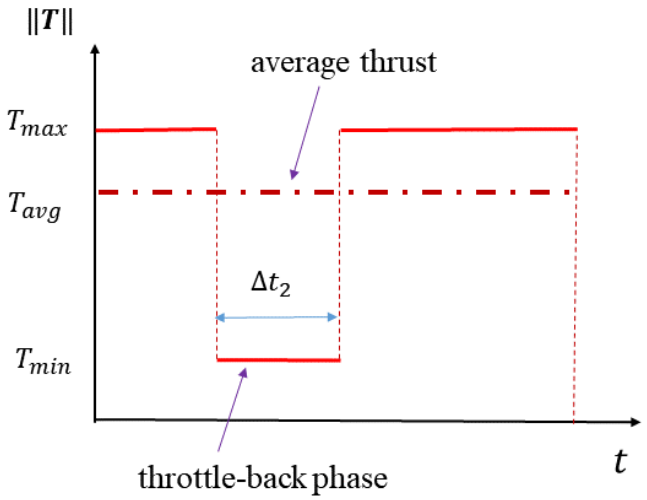


Figure 4. Fuel-optimal thrust magnitude profile

throttle-back phase allows another thrust profile that is below T_{max} to offer the equivalent thrust impulse. Figure 4 shows an average thrust profile with $T_{avg} < T_{max}$ that has the same thrust impulse with the bang-bang thrust profile. Physically, it means that the PDI condition found will allow powered descent trajectories to have thrust requirements not always on the maximum limit, and an Apollo-class powered descent guidance law will be feasible. This margin is enabled by the constraint in Eq. (43). Furthermore, using a larger value for Δ_2 provides more guaranteed margin, which in turn not only improves the feasibility of powered descent, but also increases the guidance robustness of the Apollo-class guidance laws against dispersions and modeling uncertainty (e.g., engine thrust dispersions). On the other hand, a larger lower bound on Δt_2 can increase the propellant usage in the powered descent phase because the PDI velocity is higher (when the unpowered trajectory is entry flight). In other words, we have a convenient and physically intuitive "knob" in the form of constraint (43) to tune in the A^{2}PDG method for the classical trade off between robustness and performance.

A typical value of Δ_2 that suffices for the applications in this paper is 2-5 seconds.

V. Numerical Demonstration

The A^{2}PDG method is demonstrated in this section with extensive closed-loop simulations.

A. Simulation Set-up

The vehicle model used in the simulations is that of a mid-lift-to-drag-ratio rigid-body vehicle (MRV). The MRV is an aeroshell concept under investigation for a human Mars mission. The description on the MRV and its subsystems can be found in Refs. [14–16]. The hypersonic lift-to-drag ratio of the MRV is 0.54, and it lands in a horizontal attitude. Figure 5 shows the configuration of the MRV in entry and powered descent. The entry mass of the MRV is 62,000 kg, and the ballistic coefficient of the vehicle is 379 kg/m^2.

End-to-end closed-loop simulations in a three-degree-of-freedom (3DOF) simulation environment are carried out from the entry interface (EI) at an altitude of 125 km to touchdown at the designated landing site. In its nominal configuration the MRV is fitted with a supersonic retro-propulsion (SRP) system with 8 engines, 4 mounted on each side. Each engine produces at full throttle a nominal thrust of 100,000 N, and has a specific impulse of 360 seconds. In the simulations in this paper it is assumed that the minimum throttle allowed is 20% and maximum throttle 100%. At the EI the vehicle's inertial velocity is 4700 m/s. The downrange to the landing site is 1200 km. The entry flight is from south to north.

In the entry phase the MRV is guided by the Fully Numerical Predictor-corrector Entry Guidance (FNPEG) algorithm. [17, 18] FNPEG is a fully automated and robust entry guidance algorithm applicable to vehicles of a wide range of lift-to-drag ratios. In the closed-loop 3DOF simulations in this paper FNPEG is called at 1 Hz. Only bank angle modulation is used for trajectory control, while the angle of attack is held at a constant value. The bank angle rate and acceleration are limited by 15 deg/s and 5 deg/s^2, respectively. For 6DOF simulation results for the MRV with FNPEG, the reader is referred to Refs. [15, 16]. The PDI point for each trajectory is determined by the method

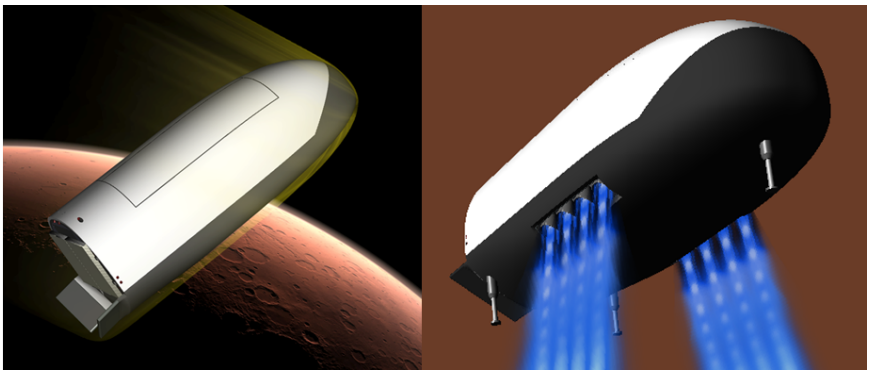


Figure 5. Mid lift-to-drag-ratio rigid-body vehicle in entry and powered descent

described in Section IV-B. During the powered descent, the tunable Apollo powered descent guidance law in Eq. (18) with $k_r = 6$ and $\boldsymbol{a}^*_{T_f} = 2\boldsymbol{g}$ (where $\boldsymbol{g}$ is the Martian gravitational acceleration) is called at a frequency of 10 Hz. For the MRV, the SRP engine thrust is assumed to be aligned with the body-normal axis (see Fig. 5). The direction of the thrust vector therefore defines the body frame orientation. In powered descent simulations, the thrust vector direction (thus the vehicle body attitude) is defined in a 3-2-1 (yaw-pitch-roll) rotation sequence by the Euler angles with respect to the North-East-Down (NED) frame at the landing site.

The targeting condition for the tunable Apollo powered descent guidance law in Eq. (18) is 12.5 m above the landing site with a descent rate of 2.5 m/s. After that the vehicle is guided by the gravity-turn guidance law in Eq. (32) to touchdown on the ground at a vertical velocity of 2.5 m/s. A 3DOF trajectory is considered a success if the miss distance to the landing site is within 50 meters.

A total of 1000 cases are run in each scenario. The dispersions include those on the EI condition, entry mass, lift and drag coefficients of the vehicle, Martian atmosphere, and SRP thrust. The dispersions are listed in Table 1. The Martian atmospheric density dispersions are provided by the Mars Global Reference Atmospheric Model (Mars GRAM 2010) [19] which generates random atmospheric density dispersions dependent on a number of parameters including altitude, location, time, and others. A parameter in Mars GRAM 2010 is *dusttau* that specifies the optical depth of background dust level which affects the atmospheric density as well. The value of *dusttau* is also dispersed uniformly between 0.1 and 0.9 in our simulations. It should be mentioned that the EI condition dispersions and SRP thrust dispersion in Table 1 are rather severe. Except for the EI condition, all uncertainties and dispersions are not known to the guidance system which uses only nominal values or nominal models. On the other hand, perfect navigation information and control system are assumed in this paper, thus the testing results represent the full capability of the guidance algorithms.

Table 1. Dispersions for Monte Carlo Simulations

Dispersion	distribution	3σ/maximum	-3σ/minimum
EI planetodetic altitude (km)	Gaussian	0.3	-0.3
EI longitude (deg)	Gaussian	0.75	-0.75
EI planetodetic latitude (deg)	Gaussian	0.75	-0.75
EI inertial velocity (m/s)	Gaussian	10.0	-10.0
EI inertial flight-path angle (deg)	Gaussian	0.3	-0.3
EI inertial azimuth (deg)	Gaussian	0.5	-0.5
EI vehicle mass (percentage)	Gaussian	1.0%	-1.0%
C_L (percentage)	Uniform	10.0%	-10.0%
C_D (percentage)	Uniform	10.0%	-10.0%
Atmospheric density	Mars GRAM 2010	Mars GRAM 2010	Mars GRAM 2010
Mars GRAM 2010 *dusttau*	Uniform	0.9	0.1
Total SRP max thrust (percentage)	Gaussian	3.0%	-3.0%

B. Evaluation and Comparison

To first demonstrate the traditional way in which Apollo powered descent guidance is deployed, the PDI condition is defined in simulations by a specified threshold of ground range to the landing site. A corresponding initial value of t_{go} must also be chosen *a priori*. As soon as this threshold is crossed, the guidance law in Eq. (18) takes over from FNPEG. Table 2 summarizes the statistics of the touchdown conditions (when the planetodetic altitude is zero) for 3 sets of the ground range threshold and initial t_{go}. The first two sets of these values are determined by trial and error so that all the Monte Carlo runs are successful. The third case has seemingly peculiar values for the PDI ground range threshold and initial t_{go}. They will be explained shortly.

It can be seen from Table 2 that in the first two cases, the MRV lands at the designated site accurately in all runs. The trend in propellant usage clearly suggests that pushing the PDI point closer to the landing site with a small t_{go} would likely further reduce the propellant consumption. Case 3 follows this observation with an even shorter PDI ground range trigger and smaller t_{go}. However, the statistics for the Case 3 (the mean and standard deviation in particular) on the miss distances unmistakably indicate that a considerable number of the runs failed to land within 50 meters of the landing site. Figure 6 shows that indeed more than 100 trajectories landed with misses significantly larger than 50 meters. The statistics on the touchdown descent rates for Case 3 in Table 2 and Fig. 7 also reveal that the large landing location misses are accompanied by relative large final velocities (up to 226 m/s). These failures were invariably caused by engine thrust saturation, a clear indication that the PDI condition and the initial t_{go} were not good choices in these cases. In fact, without help from other sources, it is not easy to find by trial and error a constant PDI ground range trigger much smaller than 15 km and the pairing initial value of t_{go} that will allow all dispersed trajectories to touchdown accurately with any variant of the family of Apollo powered descent guidance laws. This difficulty highlights the obstacles the conventional APDG approach faces.

Table 2. Statistics of touchdown conditions under tunable guidance in Eq. (18) (k_r = 6.0); PDI determined by prescribed ground range; initial t_{go} given

PDI criterion and t_{go}	parameter	mean	stdev	max	min
range=30.0 km	miss (m)	9.38E-02	7.99E-03	1.13E-01	6.77E-02
t_{go} = 130.0 s	descending rate (m/s)	2.50E+00	1.90E-03	2.50E+00	2.49E+00
	fuel usage (kg)	**1.32E+04**	3.04E+02	1.40E+04	1.23E+04
range=20.0 km	miss (m)	1.01E-01	8.38E-03	1.22E-01	7.46E-02
t_{go} = 100.0 s	descending rate (m/s)	2.50E+00	1.88E-03	2.50E+00	2.49E+00
	fuel usage (kg)	**1.17E+04**	2.62E+02	1.25E+04	1.09E+04
range=11.5 km	miss (m)	5.49E+01	2.16E+02	1.53E+03	4.34E-03
t_{go} = 52.4 s	descending rate (m/s)	1.10E+01	3.31E+01	2.26E+02	2.49E+00
	fuel usage (kg)	9.47E+03	6.19E+02	1.08E+04	5.91E+03

Now, the PDI logic developed in Section IV is used in the closed-loop simulation to determine the PDI point, and the guidance law remains unchanged - that is, we apply the A^{2}PDG method. For the same nominal mission, the PDI condition is found to be at a ground range of 11.17 km to the landing site with a velocity of 497.44 m/s and altitude of 2.42 km. The nominal altitude versus distance to landing site during the powered descent is plotted in Fig. 8. The nominal engine throttle and vehicle body pitch and yaw angles are shown in Fig. 9. The nominal engine throttle is sufficiently away from either upper bound (1.0) or lower bound (0.25), indicating a satisfactory margin. The nearly constant small throttle at the end is from the 5-second vertical landing phase under the guidance law in Eq. (32). The vehicle body Euler angles suggest that the maneuvers are mainly in the pitch plane as expected, yet there is still non-zero lateral motion due to the variations of the yaw angle (i.e., the powered descent trajectory is not exactly planar). Note that a zero pitch angle at the end means that the vehicle lands horizontally as desired (cf. Fig. 5).

Next the A^{2}PDG method is applied to guide each of the dispersed trajectories. Everything else, including the dispersions, entry and powered descent guidance algorithms, is the same as in the preceding simulations. Table 3 lists the statistics of the PDI conditions generated for the 1000 trajectories. The mean PDI ground range and powered descent time are respectively 11.5 km and 53.4 s, exactly those used in Case 3 in Table 2. However, the PDI ground range determined by the PDI logic in a particular trajectory can be as much as 13.8 km and as little as 9.46 km, and the initial t_{go} can range from 48.3 s to 56.4 s. Figure 10 illustrates the distributions of the PDI ground ranges and powered descent times. The statistics of other parameters in Table 3 also suggest that no constant threshold for any of these parameters could be a one-size-fits-all trigger for PDI.

The statistics of the touchdown conditions for the same 1000 trajectories are given in Table 4. In contrast to Case 3 in Table 2, the PDI logic in Section IV adaptively determines a favorable PDI condition for each dispersed trajectory

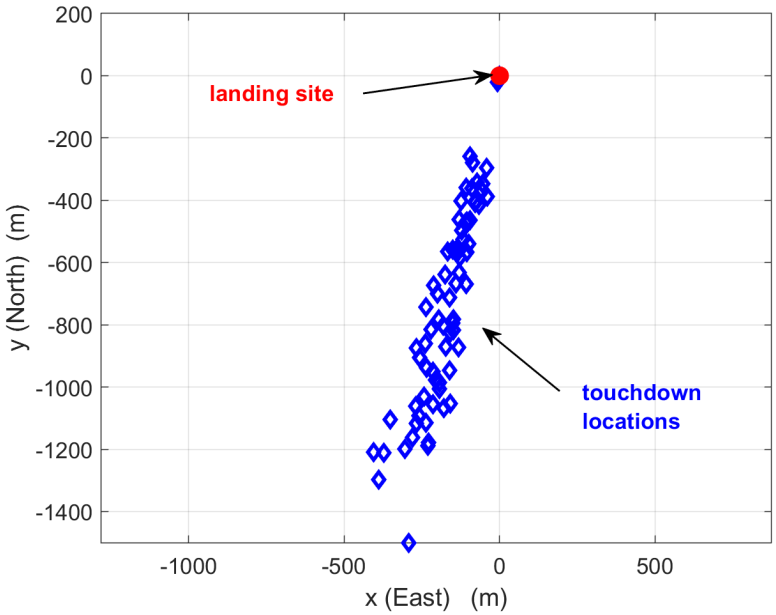


Figure 6. Touchdown locations for Case 3 in Table 2

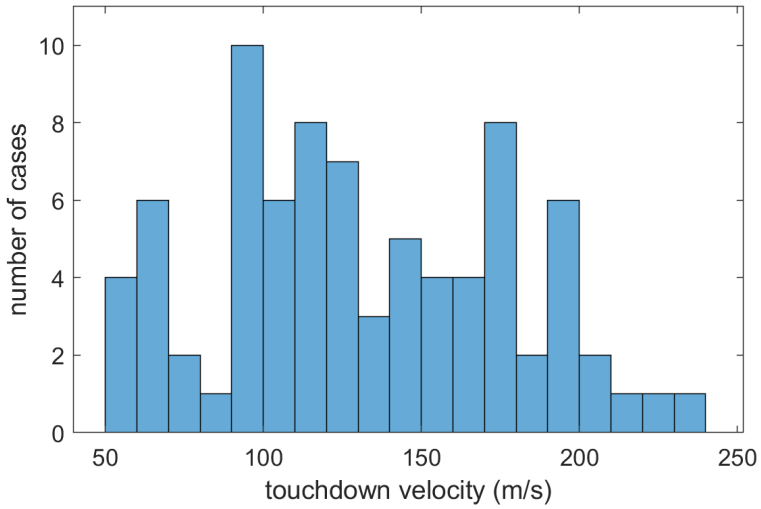


Figure 7. Instances with high touchdown velocities for Case 3 in Table 2

Table 3. Statistics of PDI conditions determined by the adaptive PDI algorithm in Section IV

parameter	mean	standard deviation	maximum	minimum
ground range (m)	1.15E+04	7.40E+02	1.38E+04	9.46E+03
powered descent time (s)	5.24E+01	1.35E+00	5.64E+01	4.83E+01
altitude (m)	2.48E+03	3.83E+02	3.38E+03	1.37E+03
velocity (m/s)	5.05E+02	1.89E+01	5.63E+02	4.53E+02
time since EI (s)	4.54E+02	1.11E+01	4.84E+02	4.19E+02

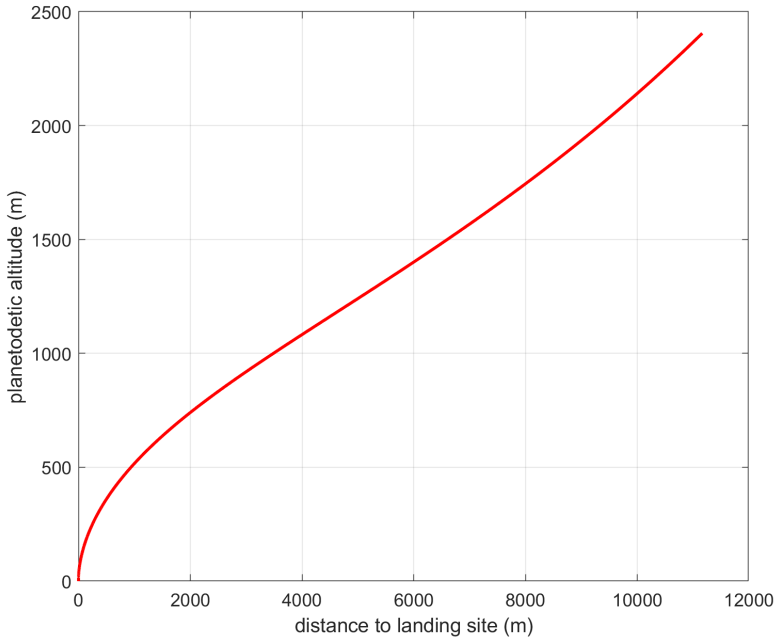


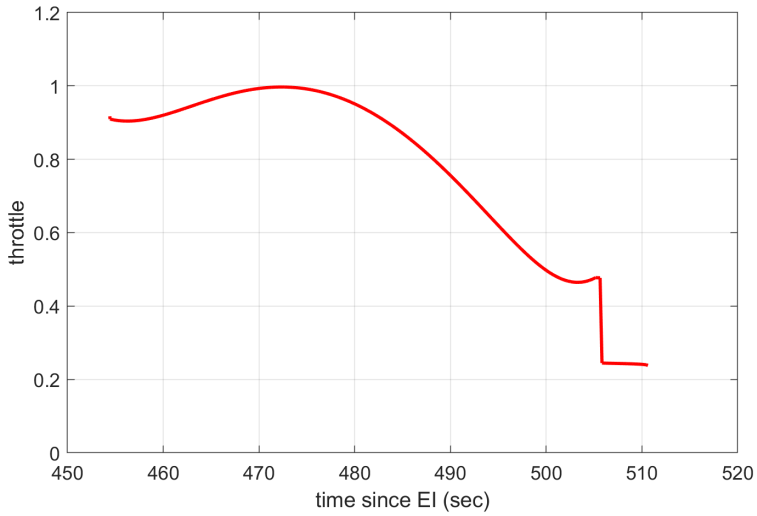
Figure 8. Nominal altitude versus ground range to landing site in powered descent

and enables successful and accurate touchdown for all the trajectories. Figure 11 depicts the landing locations of the trajectories. All the misses are within 0.15 meter. The small misses are due to the modeling uncertainties (thrust, mass, etc) in the Monte Carlo simulations and the finite guidance command update frequency. But they should be put in the context of perfect navigation and 3DOF simulations (although trajectory control rates are limited as well as control acceleration for entry). Still, the results serve to demonstrate the capability of the guidance algorithms if navigation and control systems are perfect.

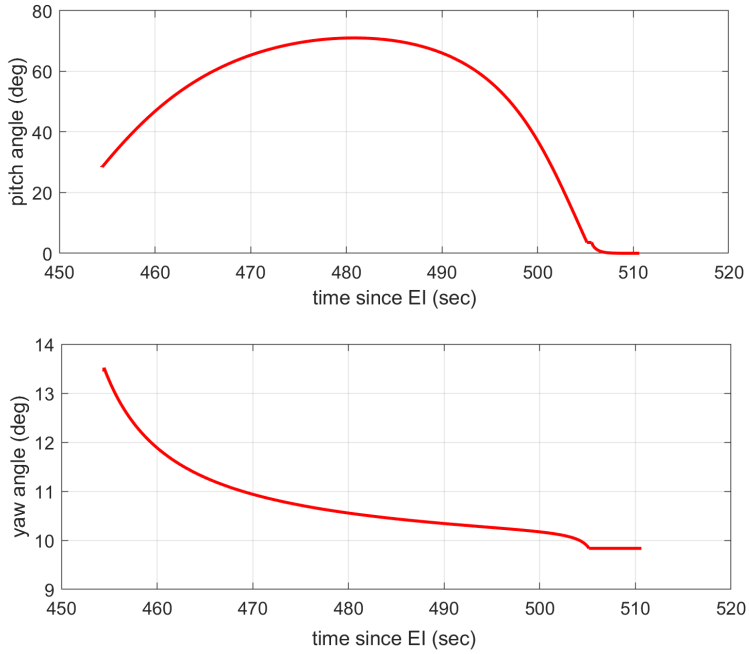
The mean fuel consumption in Table 4 is 9,490 kg. The maximum propellant usage is 10,300 kg. The histogram of the the distribution of the propellant usage of all the trajectories is plotted in Fig. 12. This mean propellant usage is 3,710 kg less than that of Case 1, and 2,200 kg less than Case 2 in Table 2. These are significant differences (39.1% and 23.2% reduction, respectively), all just due to the difference in when to start the powered descent phase.

For any other values of $k_r \in [6, 12]$ in the guidance law in Eq. (18), all the trajectories land just as accurately, with different slopes (see Fig. 1). For instance, when $k_r = 12.0$, that is, the APDG law is used, the trajectory has a steeper slope toward the end, thus the transition to the final vertical landing phase is much gentler than $k_r = 6$. But the mean propellant usage is about 450 kg more than that with $k_r = 6.0$ in Table 4.

In comparison, if the fuel-optimal guidance algorithm UPG is used for closed-loop guidance as in Ref. [12], the mean propellant consumption for the same 1,000 runs is 9,210 kg, only 280 kg less than the mean propellant usage in Table 4 (a difference of 2.9%). Therefore the A^2PDG method is able to yield a near-optimal propellant performance. On the other hand, the guidance law in Eq. (18) is dramatically simpler than the UPG algorithm. And equally importantly, the guidance law has a much higher technology readiness level than UPG because it is based on the flight-proven technology of APDG.



a) Engine throttle profile

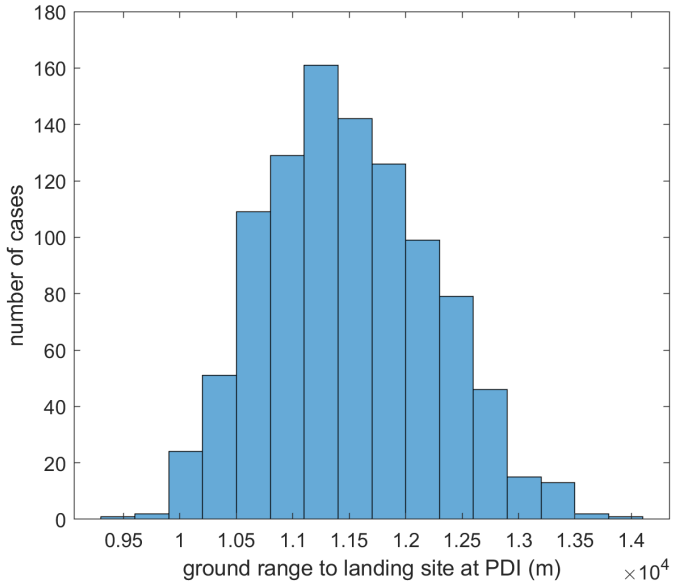


b) Euler angles of the vehicle

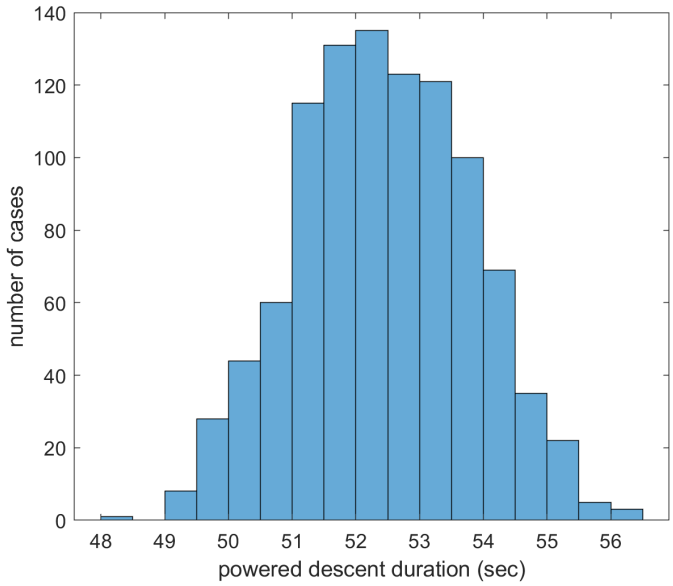
Figure 9. Nominal profiles in powered descent

Table 4. Statistics of touchdown conditions with the A^{2}PDG method ($k_r = 6.0$) with the adaptive PDI algorithm in Section IV

parameter	mean	standard deviation	maximum	minimum
miss (m)	5.68E-02	2.28E-02	1.47E-01	9.98E-03
descending rate (m/s)	2.50E+00	1.84E-03	2.50E+00	2.49E+00
flight path angle (deg)	-7.90E+01	4.23E-01	-7.79E+01	-8.01E+01
pitch angle (deg)	6.31E-03	1.79E-04	6.91E-03	5.76E-03
propellant usage (kg)	**9.49E+03**	2.84E+02	1.03E+04	8.64E+03



a) Distribution of PDI ground ranges



b) Distribution of powered descent times

Figure 10. PDI condition distributions by the adaptive PDI algorithm in Section IV

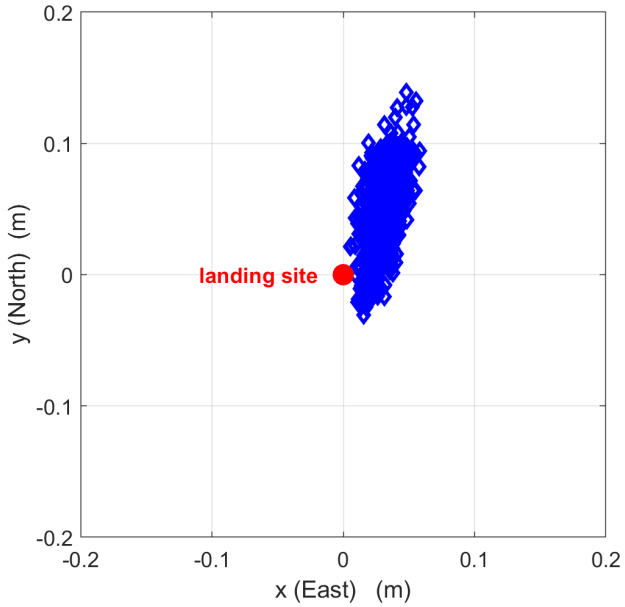


Figure 11. Landing locations under A^{2}PDG with the adaptive PDI algorithm in Section IV

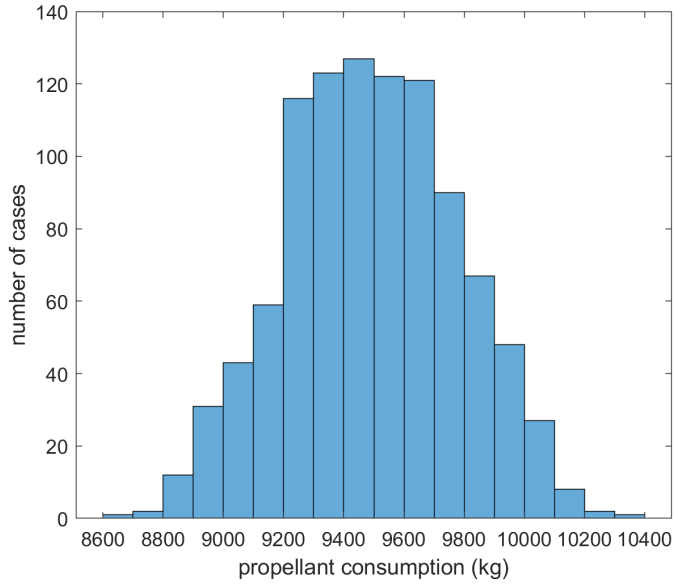


Figure 12. Propellant usages of the 1000 dispersed trajectories under A^{2}PDG with the adaptive PDI algorithm in Section IV

VI. Conclusions

In aerospace guidance and control, especially for space flight, great premium is placed on technologies that have been flown before. The venerable Apollo powered descent guidance (APDG) approach is such a technology that landed the Lunar Module on the Moon for all successful lunar landing missions. Yet, the needs 50 years later for autonomous operations, higher performance, and diverse applications have amplified some of the weaknesses in the APDG. In this work we develop a methodology called Augmented Apollo Powered Descent Guidance (A^2PDG) to modernize the legacy technology of APDG with the latest research advances. A family of variants of the Apollo powered descent guidance law are derived from a perspective of implicit guidance. The classical APDG law and well-known Apollo E-guidance law are shown to be two special cases of this family. Any member of the tunable guidance laws is defined by the value of an adjustable gain, and shares the similar simplicity, reliability, and predictability of the APDG approach. At the front end of A^2PDG, the guidance command is generated by such a tunable Apollo powered descent guidance law. Behind the scenes, the guidance system is supported by a novel online powered descent initiation (PDI) logic based on an advanced propellant-optimal rocket descent algorithm. This addition addresses effectively the long-standing challenge of determining online a good PDI condition and initial time-to-go when applying APDG, and allows the guidance system to provide a nearly fuel-optimal performance. The latter (fuel-optimality) is critically important for a high-mass Mars mission. The A^2PDG framework inherits the strengths of the flight-proven legacy technology of APDG, yet enables the powered descent guidance system to be fully autonomous, propellant efficient, and adaptive to different flight conditions and even in-flight change of landing site. Extensive end-to-end closed-loop Monte Carlo simulations for a human Mars mission have provided compelling evidence to the potential of the A^2PDG methodology.

Acknowledgments

This research was supported by NASA Cooperative Agreement NNX16AL35A. The author thanks the anonymous reviewers for this paper whose comments and suggestions have helped improve the presentation.

References

[1] Klumpp, A. R., "A Manually Retargeted Automatic Descent and Landing System for LEM," *Proceedings of AIAA/JACC Guidance and Control Confernece*, Seattle, WA, 1966, pp. 713–727. doi:6.1966-1863.

[2] Bennett, F. V., "Apollo Lunar Descent and Ascent Trajectories," NASA TM X-58040, 1970.

[3] Klumpp, A. R., "Apollo Lunar Descent Guidance," *Automatica*, Vol. 10, No. 2, 1974, pp. 133–146. doi:10.1016/0005-1098(74)90019-3.

[4] Cherry, G. W., "A General, Explicit, Optimizing Guidance Law for Rocket-Propelled Spaceflight," AIAA Paper 64-638, 1964. doi:10.2514/6.1964-638.

[5] D'Souza, C. S., "An Optimal Guidance Law for Planetary Landing," AIAA Paper 97-3709, 1997. doi:10.2514/6.1997-3709.

[6] Carmen, G. L., Ives, D. G., and Geller, D. K., "Apollo Derived Mars Precision Lander Guidance," AIAA Paper 98-4570, 1998. doi:10.2514/6.1998-4570.

[7] Wong, E., Singh, G., and Masciarelli, J., "Guidance and Control Design for Hazard Avoidance and Safe Landing on Mars," *Journal of Spacecraft and Rockets*, Vol. 43, No. 2, 2006, pp. 378–384. doi:10.2514/1.19220.

[8] Sostaric, R. and Rea, J., "Powered Descent Guidance Methods For The Moon and Mars," AIAA Paper 05-6287, 2007. doi:10.2514/6.2005-6287.

[9] Steinfeldt, B. A., Grant, M. J., Matz, D. M., Braun, R. D., and Barton, G. H., "Guidance, Navigation, and Control Technology System Trades for Mars Pinpoint Landing," *Journal of Spacecraft and Rockets*, Vol. 47, No. 1, 2010, pp. 188–198. doi:10.2514/1.45779.

[10] Gerth, I. and Mooij, E., "Guidance for Autonomous Precision Landing on Atmosphereless Bodies," AIAA Paper 2014-0088, 2014. doi:10.2514/6.2014-0088.

[11] Lu, P., "Fuel-Optimal and Apollo Powered Descent Guiance Compared for High-Mass Mars Mission," *Advances in the Astronautical Sciences, Vol. 164*, 2018, pp. 299–314.

[12] Lu, P., "Propellant-Optimal Powered Descent Guidance," *Journal of Guidance, Control, and Dynamics*, Vol. 41, No. 4, 2018, pp. 813–826.
doi:10.2514/1.G003243.

[13] Lu, P., Sostaric, R. R., and Mendeck, G. F., "Adaptive Powered Descent Initiation and Fuel-Optimal Guidance for Mars Applications," AIAA Paper 2018-0616, 2018.
doi:10.2514/6.2018-0616.

[14] Cerimele, C., Robertson, E., Sostaric, R., Campbell, C., R. P., Matz, D., Johnson, B., and Stachowiak, S., "A Rigid Mid Lift-to-Drag Ratio Approach to Human Mars Entry, Descent, and Landing," AIAA Paper 2017-1898, 2017.
doi:10.2514/6.2017-1898.

[15] Breanna, J. J., Cerimele, C. J., Stachowiak, S. J., Sostaric, R. R., Matz, D. A., and Lu, P., "Mid-Lift-to-Drag Ratio Rigid Vehicle Control System Design and Simulation for Human Mars Entry," AIAA Paper 2018-0615, 2018.
doi:10.2514/6.2018-0615.

[16] Breanna, J. J., Braden, E. M., Sostaric, R. R., Cerimele, C. J., and Lu, P., "Entry, Descent, and Landing Performance for a Mid-Lift-to-Drag Ratio Vehicle at Mars," *Advances in the Astronautical Sciences, Vol. 164*, 2018, pp. 285–297.

[17] Lu, P., "Entry Guidance: a Unified Method," *Journal of Guidance, Control, and Dynamics*, Vol. 37, No. 3, 2014, pp. 713–728.
doi:10.2514/1.62605.

[18] Lu, P., Brunner, C., Stachowiak, S., Mendeck, G., Tigges, M., and Cerimele, C., "Verification of a Fully Numerical Entry Guidance Algorithm," *Journal of Guidance, Control, and Dynamics*, Vol. 40, No. 2, 2017, pp. 230–247.
doi:10.2514/1.G000327.

[19] Justh, H. L., *Mars Global Reference Atmospheric Model 2010 Version: Users Guide*, NASA/TM-2014-217499, 2014.